A Data-Driven Analysis of Team Performance in the Premier League and La Liga (2018-2023)

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1. Introduction

The purpose of this project is to analyze the difference between teams' performance in the Premier League and La Liga over a period of five years (2018-2023). The analysis focuses on key metrics such as Goals scored, Expected Goals, Efficiency and contextual factors like home and away display. The main objective is to evaluate team performance not only in terms of chance creation but also their finishing ability between these two leagues during this period.

2. Data & Methodology

The dataset consists of match-level data from Fbref and TransferMarkt which are known as reliable providers of football statistics. This includes key variables like Goals and xG for both home and away teams in Premier League and La Liga. Data was preprocessed by using *Python* in which goals and expected goals were extracted. Incomplete data from the 2023-2024 season was excluded to ensure consistency. A season label was assigned for each match, matches played between August and December were assigned to the current season (e.g., August 2019 → 2019-2020), while matches played between January and May were assigned to the latter part of the same season (e.g., April 2019 → 2018-2019). Additionally, an important metric, "Efficiency" was calculated by using the difference between each teams' goals scored and xG (Goals – Expected Goal) representing their finishing of chances. The final dataset consists of 7600 rows including 3800 matches from Premier League and 3800 from La Liga. Finally processed data was visualized by using *Tableau* to identify patterns and trends across these leagues.

3. Analysis

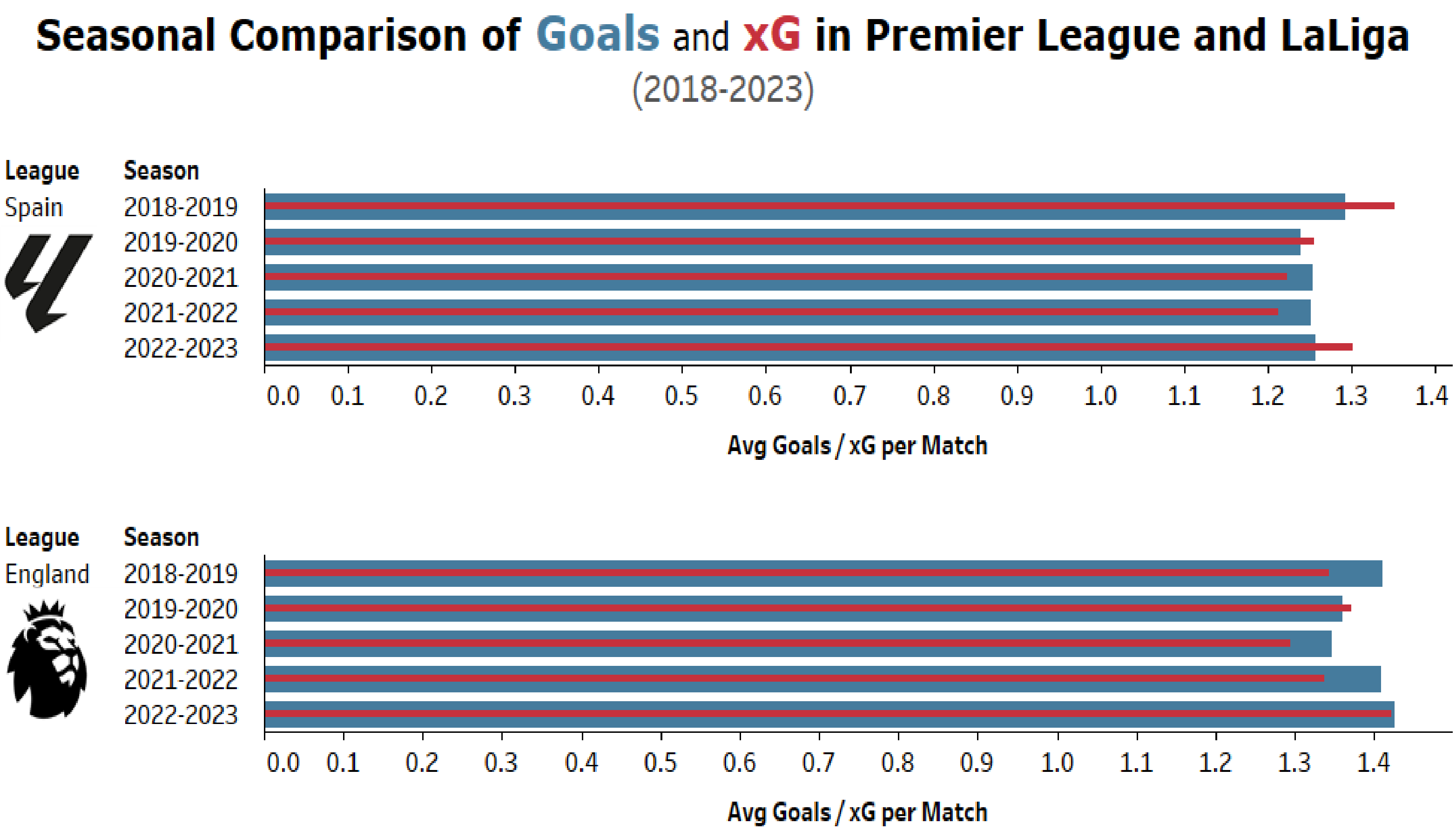
This section of the article evaluates teams' performance based on the defined metrics in Premier League and La Liga. The analysis has several parts; league-level trends, the relationship between goals and expected goal (xG), home and away performance and efficiency pattern over the time. The aim is to identify key differences between these two leagues.

3.1 League-Level Attacking Trends

The results indicate that both leagues remained roughly the same in terms of goal scoring, however, their expected goals show greater variation across seasons.

Between 2018 and 2021, the number of goals for Premier League saw a moderate decline but from 2021 onwards it grew substantially. Additionally, the gap between goals and xG provides insight into finishing efficiency, Premier League teams are more effective in converting their chances as the number of goals exceeded their xG in several seasons. By contrast, in La Liga the number of goals stayed approximately the same during this period but their scoring ability is not as efficient as Premier League. One factor that we can take into consideration is that Premier League spent more money for players on average (€2.2B) during this period than La Liga (€1B), so the quality of their League is relatively higher compared with their Spanish peer.

Overall, these findings suggest that while both leagues maintain consistent attacking output, Premier League tends to exhibit a higher offensive intensity.



3.2 Goal vs Expected Goal: Finishing Efficiency

The scatter plots were used to evaluate the relationship between goals scored and expected goals (xG) in both Premier League and La Liga. Since xG measures the quality of scoring opportunities, comparing it with actual goals helps identify how efficiently teams convert their chances.

Overall, the results show a strong positive relationship between goals and xG in both leagues. Teams with higher expected goals generally scored more goals, suggesting that xG is a reliable indicator of attacking performance. However, differences between the two leagues become more visible when examining the distribution of teams and their finishing efficiency.

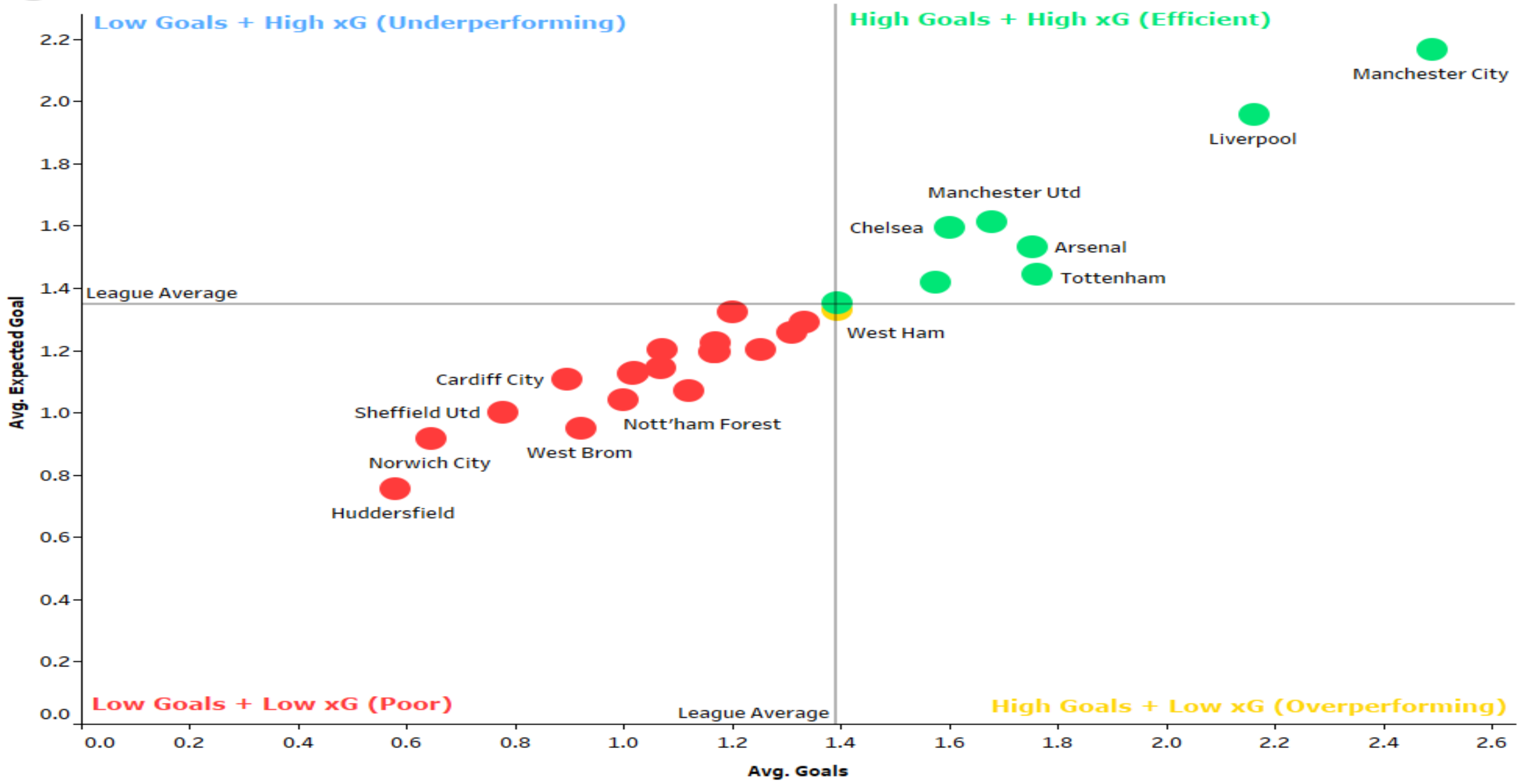
In the Premier League, the distribution of teams is wider, indicating greater variability in attacking performance. Several top teams not only generated high xG values but also scored more goals than expected, reflecting stronger finishing quality and attacking intensity. This may suggest that Premier League teams are generally more aggressive and efficient in converting chances.

By contrast, La Liga shows a more compact distribution, with most teams positioned closer to the league average. Although elite teams such as Barcelona and Real Madrid maintained strong attacking output, fewer teams significantly exceeded their xG values. This suggests a more balanced but less variable attacking structure across the league.

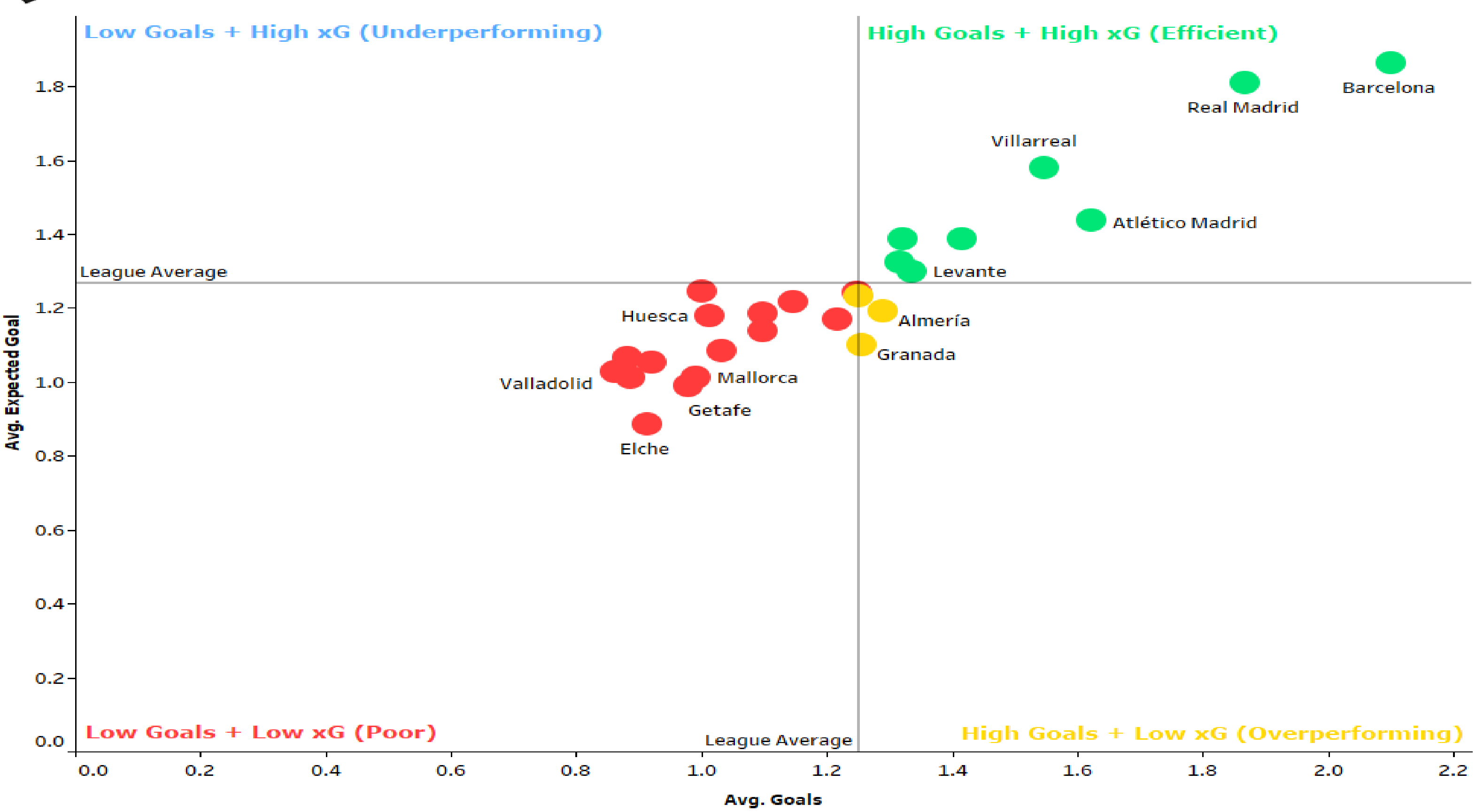
Interestingly, the underperforming quadrant (high xG but low goals) is largely unoccupied in both leagues. Since the analysis is based on five-season averages, short-term finishing inefficiencies are reduced, causing most teams to move closer to their expected performance levels over time.

Overall, the analysis demonstrates that while both leagues maintain a strong relationship between chance creation and scoring, the Premier League displays greater attacking variability and slightly stronger finishing efficiency.

Premier League: Goals vs xG
(2018-2023)



LaLiga: Goals vs xG
(2018-2023)

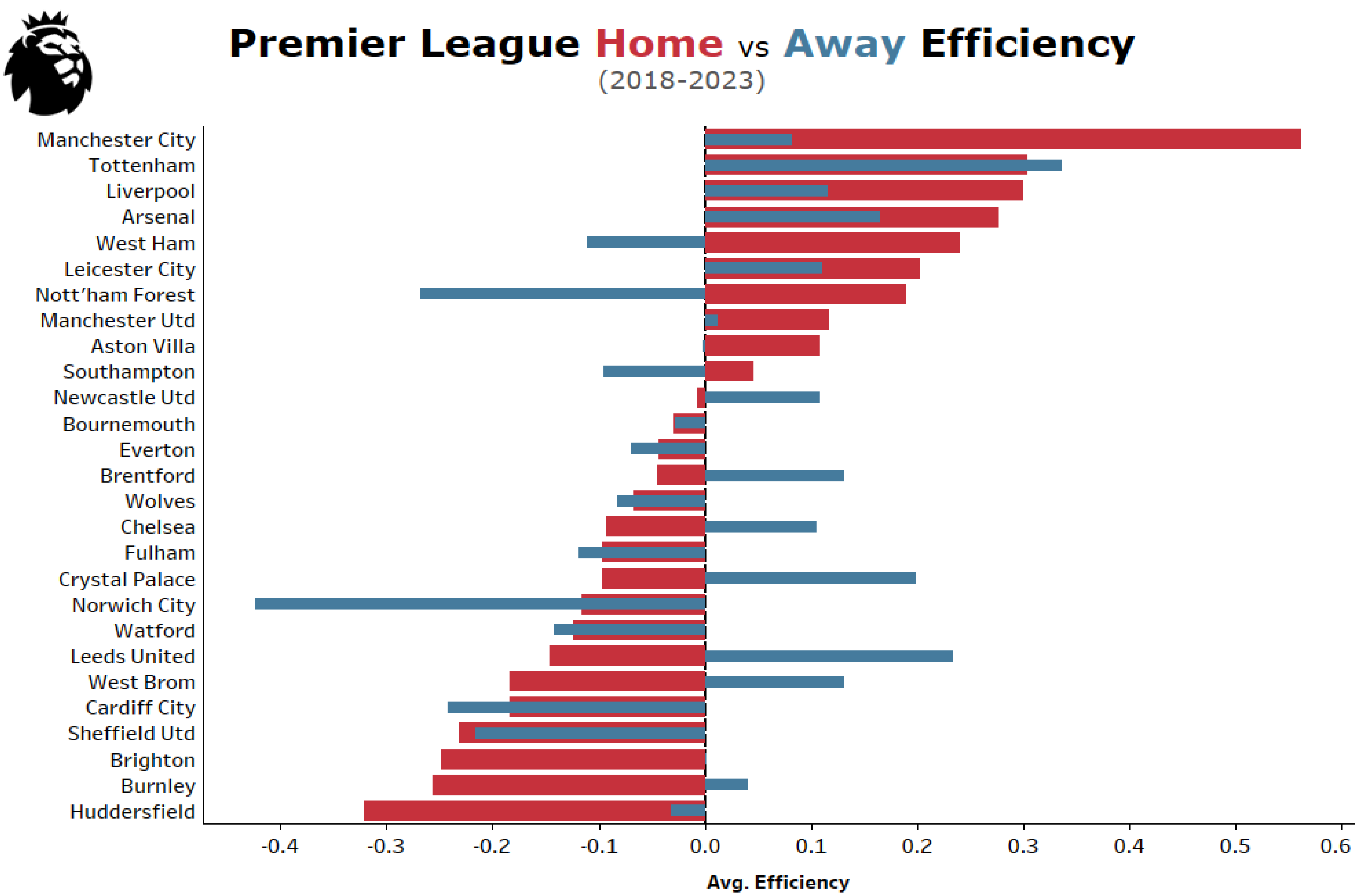


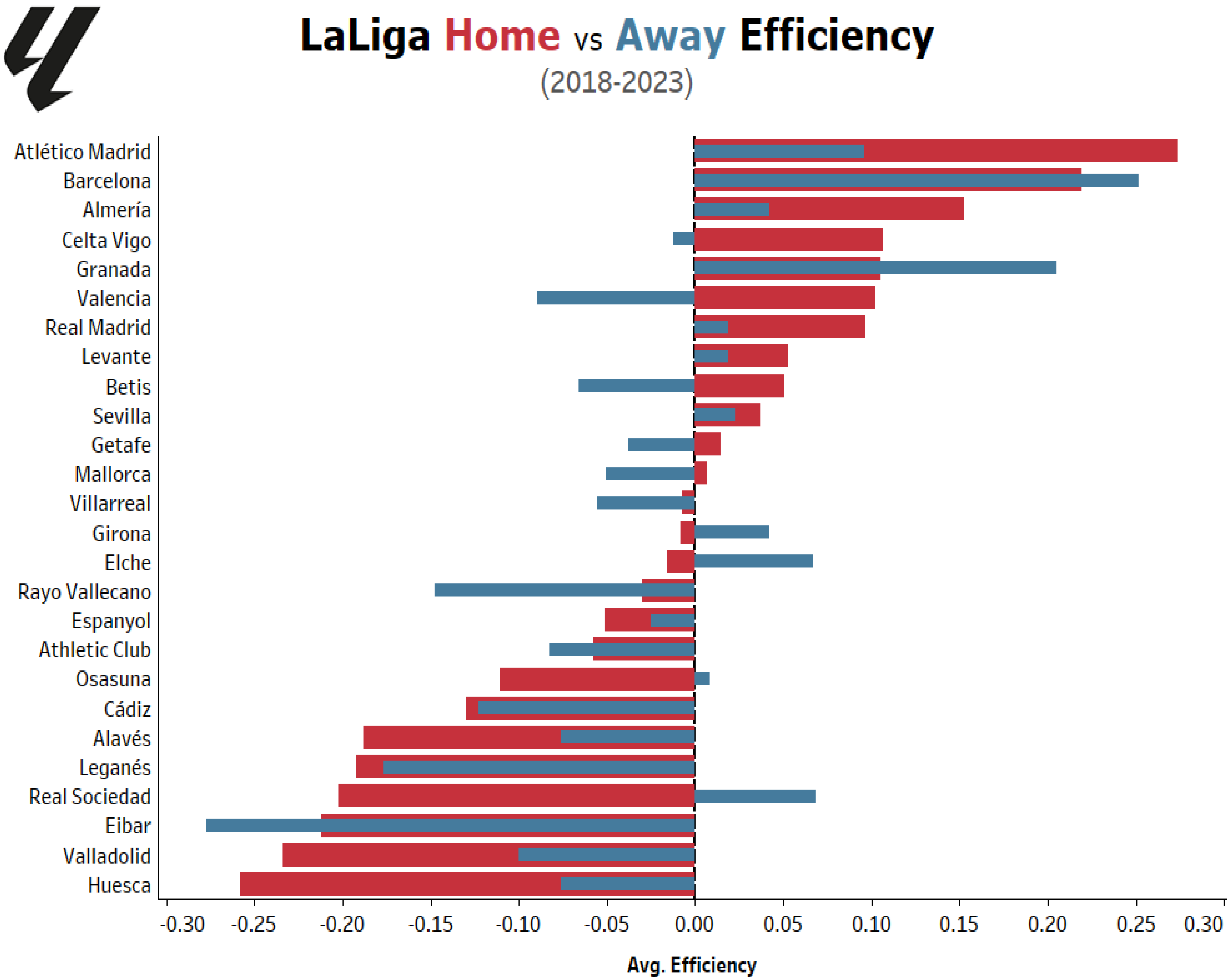
3.3 Contextual Efficiency: Home vs Away Performance

These graphs reveal the difference in teams' performance depending on match location.

In La Liga most teams have higher finishing in their own stadiums, but in Premier League a few teams showed their tactical dominance in home matches, this may suggest greater adaptability in away performances among Premier League teams.

Several top teams like Manchester City, Liverpool, Barcelona, Real Madrid and etc. displayed relatively balanced efficiency between home and away matches and it shows their tactical consistency across different environments, even some of them like Tottenham or Atletico Madrid had better finishing in away matches than home matches. On the other hand, teams such as Manchester United and Celta Vigo appear to rely more heavily on home performances, suggesting lower tactical adaptability in away matches. And there are lower-performing teams like Burnley and Valladolid that had a poor finishing, less chance creation and tactical weaknesses no matter where they play. Since the analysis is based on five season average, the results reflect a long-term pattern.




LaLiga Home vs Away Efficiency
(2018-2023)
Atlético Madrid
Barcelona
Almería
Celta Vigo
Granada
Valencia
Real Madrid
Levante
Betis
Sevilla
Getafe
Mallorca
Villarreal
Girona
Elche
Rayo Vallecano
Espanyol
Athletic Club
Osasuna
Cádiz
Alavés
Leganés
Real Sociedad
Eibar
Valladolid
Huesca
-0.30 -0.25 -0.20 -0.15 -0.10 -0.05 0.00 0.05 0.10 0.15 0.20 0.25 0.30
Avg. Efficiency

3.4 Seasonal Efficiency Trends

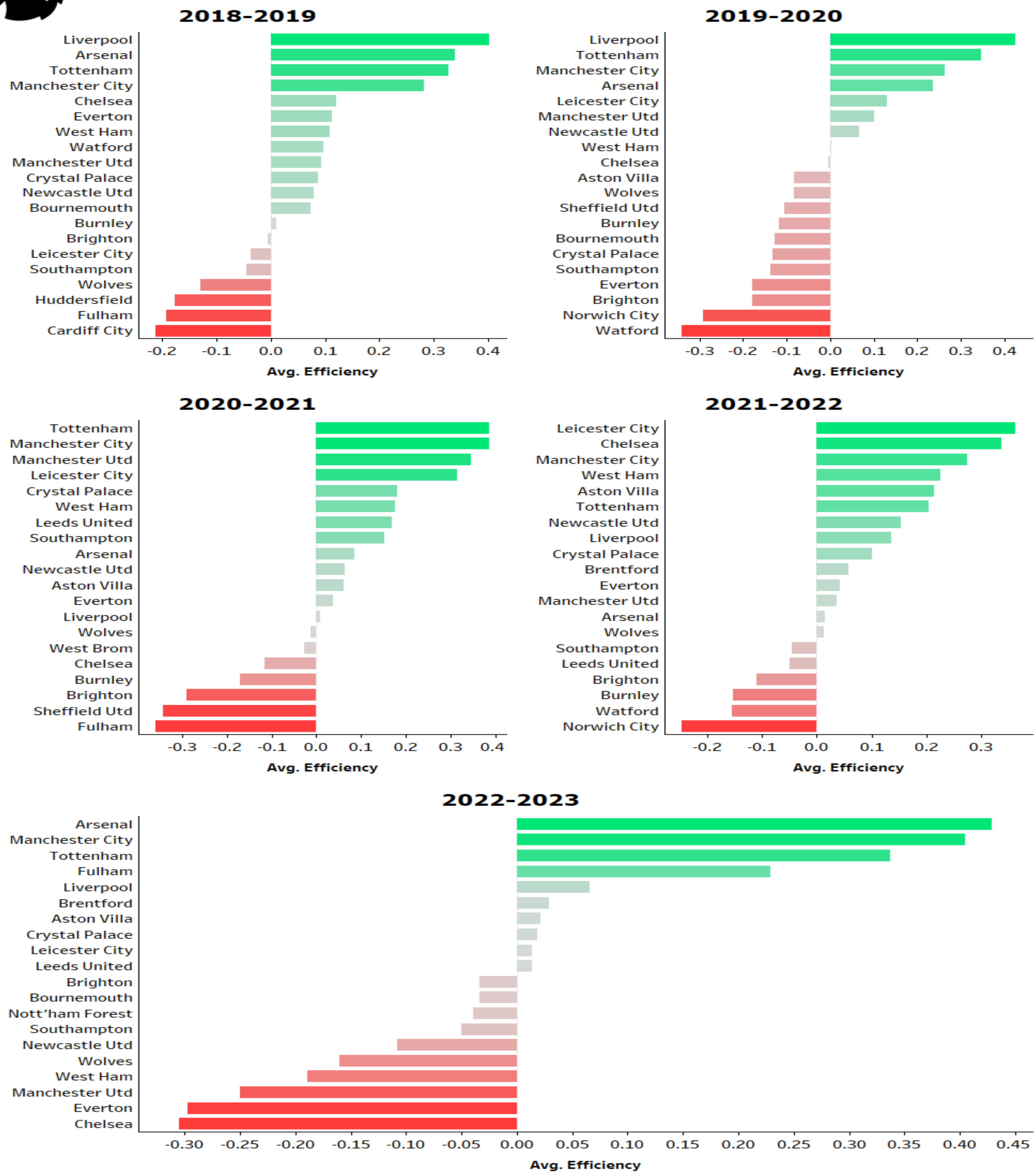
These charts illustrate the figure for teams' finishing in Premier League and La Liga. It is noticeable that this metric relies on players' quality, teams' consistency and their tactical approach during a season.

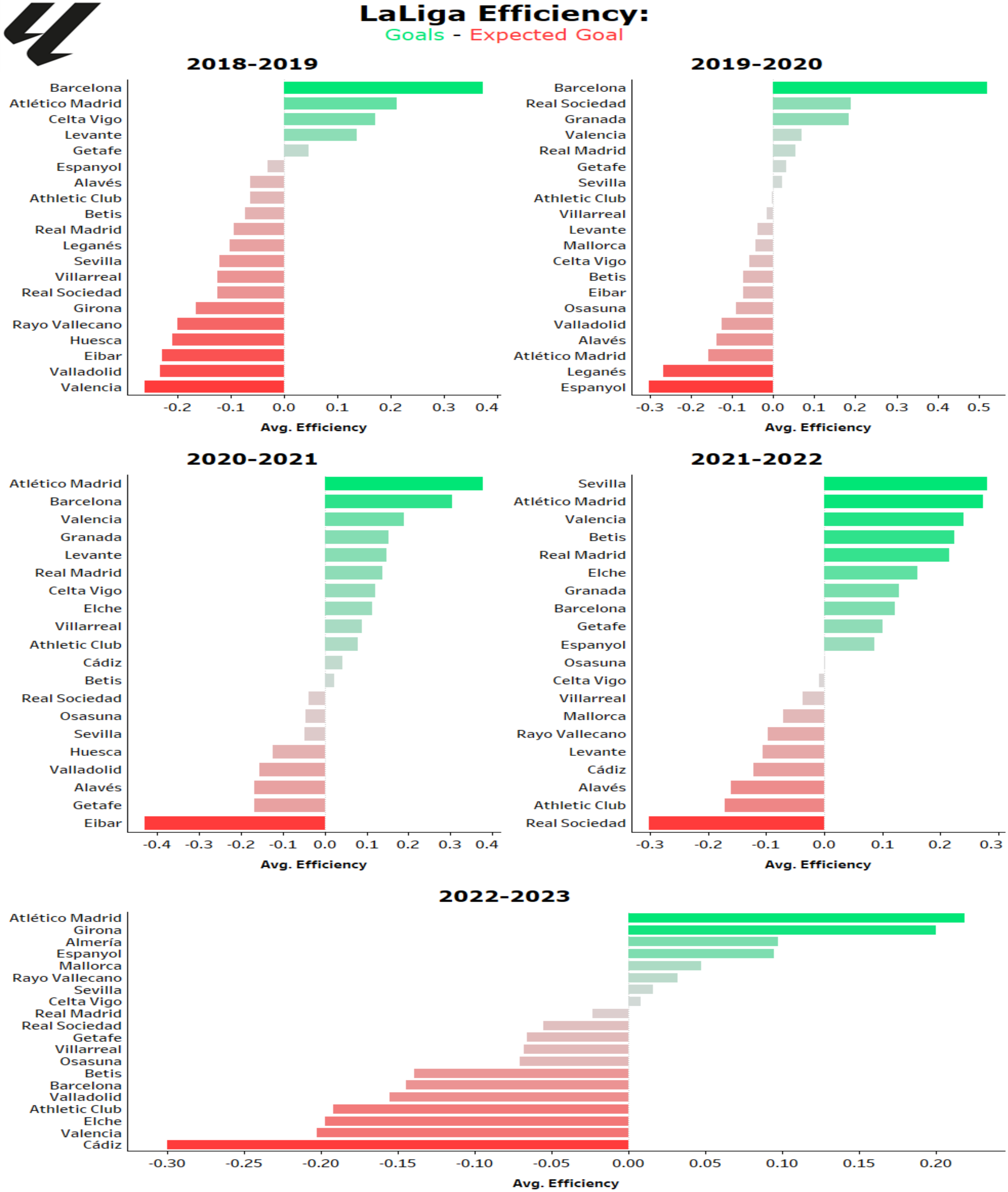
Teams like Liverpool or Atletico Madrid were consistently efficient in this period; it shows their squad quality, stable attacking structure and finishing quality. It is considerable that these efficient teams preserve their main players during this time, that help them in boosting their players' chemistry. On the other hand, there are a few teams that under/over performed extremely, for example, Barcelona recorded one of the highest efficiency values during the 2019–2020 season despite not winning La Liga. This indicates that strong finishing performance alone is not sufficient to guarantee overall league success, as factors such as defensive performance, consistency, and tactical balance also play significant roles. Conversely, Barcelona recorded a negative efficiency value during the 2022–2023 season despite winning La Liga. This suggests that overall team success is not solely dependent on finishing overperformance, but can also be driven by defensive stability, tactical control, and sustained chance creation throughout the season. Similarly, Leicester City saw the highest number of efficiency during 2021-2022 season, although they didn't win the Premier League but they managed to win the FA Cup that season, their exceptional finishing during that season help them to achieve a trophy in a cup competition. Another notable observation is the decline in Barcelona's efficiency following the 2020–2021 season, which occurred with the departure of Lionel Messi. As one of the most influential attacking players in football history, Messi's exit may have had a significant impact on Barcelona's finishing quality and attacking efficiency in subsequent seasons.

Some teams like Chelsea saw a substantial decline between 2021-2022 and 2022-2023 season. One of the most important factors that has impact in these observations is that their managerial staff completely changed after 20 years. In contrast, Arsenal under Mikel Arteta experienced a significant improvement during this period, suggesting a successful tactical evolution that brought the club close to winning the Premier League in 2022-2023 season. An interesting pattern can be observed during the 2019–2020 season, where several teams in both leagues recorded negative efficiency values. One possible explanation could be the disruption caused by the COVID-19 pandemic, which led to league interruptions, schedule congestion, and matches being played without spectators. These factors may have influenced player performance and finishing consistency during that period.

Overall, the results highlight the dynamic nature of attacking efficiency and demonstrate that team finishing performance can vary considerably across seasons.

Premier League Efficiency:
Goals - Expected Goal




LaLiga Efficiency:
Goals - Expected Goal
2018-2019
Barcelona
Atlético Madrid
Celta Vigo
Levante
Getafe
Espanyol
Alavés
Athletic Club
Betis
Real Madrid
Leganés
Sevilla
Villarreal
Real Sociedad
Girona
Rayo Vallecano
Huesca
Eibar
Valladolid
Valencia
-0.2 -0.1 0.0 0.1 0.2 0.3 0.4
Avg. Efficiency
2019-2020
Barcelona
Real Sociedad
Granada
Valencia
Real Madrid
Getafe
Sevilla
Athletic Club
Villarreal
Levante
Mallorca
Celta Vigo
Betis
Eibar
Osasuna
Valladolid
Alavés
Atlético Madrid
Leganés
Espanyol
-0.3 -0.2 -0.1 0.0 0.1 0.2 0.3 0.4 0.5
Avg. Efficiency
2020-2021
Atlético Madrid
Barcelona
Valencia
Granada
Levante
Real Madrid
Celta Vigo
Elche
Villarreal
Athletic Club
Cádiz
Betis
Real Sociedad
Osasuna
Sevilla
Huesca
Valladolid
Alavés
Getafe
Eibar
-0.4 -0.3 -0.2 -0.1 0.0 0.1 0.2 0.3 0.4
Avg. Efficiency
2021-2022
Sevilla
Atlético Madrid
Valencia
Betis
Real Madrid
Elche
Granada
Barcelona
Getafe
Espanyol
Osasuna
Celta Vigo
Villarreal
Mallorca
Rayo Vallecano
Levante
Cádiz
Alavés
Athletic Club
Real Sociedad
-0.3 -0.2 -0.1 0.0 0.1 0.2 0.3
Avg. Efficiency
2022-2023
Atlético Madrid
Girona
Almería
Espanyol
Mallorca
Rayo Vallecano
Sevilla
Celta Vigo
Real Madrid
Real Sociedad
Getafe
Villarreal
Osasuna
Betis
Barcelona
Valladolid
Athletic Club
Elche
Valencia
Cádiz
-0.30 -0.25 -0.20 -0.15 -0.10 -0.05 0.00 0.05 0.10 0.15 0.20
Avg. Efficiency

4. Conclusion

This study analyzed and compared team performance in the Premier League and La Liga between 2018 and 2023 using metrics such as goals scored, expected goals (xG), and efficiency. The objective was to evaluate attacking performance, finishing ability, and the impact of contextual factors such as home and away matches. The results indicate that the Premier League generally demonstrated higher attacking intensity and greater variability in performance compared to La Liga. In both leagues, a strong relationship between goals and xG was observed, suggesting that expected goals provide a reliable indicator of attacking output. However, efficiency values fluctuated considerably across seasons, highlighting the unstable nature of finishing performance over time. The analysis also showed that home advantage plays an important role in team efficiency, as most teams performed more effectively in home matches. Additionally, the study demonstrated that strong finishing performance alone does not necessarily guarantee league success, as tactical structure, defensive stability, and consistency also contribute significantly to overall performance**.** Future studies could expand this analysis by combining defensive metrics.

5. References

FBref. https://fbref.com

Transfermarkt. https://www.transfermarkt.com

Impact of COVID-19 on Premier League Performance. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0332627